

Working with Binary

Computers deal with data in binary (store, process, transfer)

Let's get started with decimal numbers like 1,2,10,50, etc.

1. Decimal to Binary Representation

Example: Number 1250 in **decimal number system** [aka Denary]

Base 10 (decimal) number system: Digits are 0,1,2,3,4,5,6,7,8 and 9

10^3	10^2	10^1	10^0	Place Value (Decimal)
1	2	5	0	Multiply Digit with its place value
$1 \times 10^3 = 1000$	$2 \times 10^2 = 100$	$5 \times 10^1 = 10$	$0 \times 10^0 = 1$	Value = $1000 + 200 + 50 + 0 = 1250$

Thus, the number 1250 evaluates to **ONE THOUSAND TWO HUNDRED FIFTY**

Let us look at another number 123 in decimal system.

	10^2	10^1	10^0	Place Value (Decimal)
	1	2	3	Multiply Digit with its place value
	$1 \times 10^2 = 100$	$2 \times 10^1 = 10$	$3 \times 10^0 = 1$	Value = $100 + 20 + 3 = 123$

Thus, the number 123 evaluates to **ONE HUNDRED TWENTY THREE**

Now let us see how the same number 123 can be represented in **Binary Number System**.

Earlier we have seen that In Base 10 (decimal) number system: Digits are **0,1,2,3,4,5,6,7,8 and 9**

However, in Binary system there are only two digits: **1 and 0**.

Thus, it is Base 2.

So, any data we want to represent in binary would be comprising of a 1 and 0 only.

For example: 1 0 1 1

So, let us see what it evaluates to in decimal.

2^3	2^2	2^1	2^0	Place Value (Binary)
1	0	1	1	Multiply Digit with its place value

$1 \cdot 2^3 =$ 8	$0 \cdot 2^2 =$ 0	$1 \cdot 2^1 =$ 2	$1 \cdot 2^0 =$ 1	Value = 64 + 32 + 16 + 8 + 0 + 2 + 1 = 11
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Thus, the binary **1 0 1 1** (ONE ZERO ONE ONE) evaluates to 11 (**ELEVEN**) in decimal system.

Let us do some decimal to binary conversion for a number **123**.

Here are the steps to convert from decimal to binary.

1. Start with the Decimal Number: Begin with the decimal number you want to convert to binary.
2. Divide by 2: Divide the decimal number by 2.
3. Note the Remainder: Note the remainder (either 0 or 1).
4. Write Down the Remainder: Write down the remainder as the least significant bit (LSB) of your binary representation.
5. Update the Quotient: Update the quotient to be the result of the division in step 2.
6. Repeat the Process: Repeat steps 2-5 until the quotient becomes 0.
7. Read the Binary Representation: The binary representation is read by reading the remainders from bottom to top.

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Step 1: 123 / 2 = 61 with remainder 1  --> LSB: 1
Step 2: 61 / 2 = 30 with remainder 1  --> Next bit: 1
Step 3: 30 / 2 = 15 with remainder 0  --> Next bit: 0
Step 4: 15 / 2 = 7 with remainder 1  --> Next bit: 1
Step 5: 7 / 2 = 3 with remainder 1  --> Next bit: 1
Step 6: 3 / 2 = 1 with remainder 1  --> Next bit: 1
Step 7: 1 / 2 = 0 with remainder 1  --> MSB: 1 (last bit)

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Binary representation: **1111011**

Binary representation: **1 1 1 1 0 1 1**

$(123)_{10} = (1 1 1 1 0 1 1)_2$

$(123)_{10} = (0 1 1 1 1 0 1 1)_2$ [8-bit format]

Base 2 (Binary) number system: Digits are 0 and 1.

Convert back from binary to decimal.

$$(01111011)_2 = (?)_{10}$$

2^7	2^6	2^5	2^4	2^3	2^2	2^1	2^0	Place Value (Binary)
0	1	1	1	1	0	1	1	Multiply Digit with its place value
0	$1 \times 2^6 =$ 64	$1 \times 2^5 =$ 32	$1 \times 2^4 =$ 16	$1 \times 2^3 =$ 8	$0 \times 2^2 =$ 0	$1 \times 2^1 =$ 2	$1 \times 2^0 =$ 1	Value = 64 + 32 + 16 + 8 + 0 + 2 + 1 = 123

$$(01111011)_2 = (123)_{10}$$

How are negative numbers represented in binary form? Next, we discuss a method known as **2's Complement to do so.**

2. Representing negative numbers using 2's complement.

Let us see how -123 (remember earlier we converted 123) can be represented in binary format using 2's Complement method.

Here are the steps to convert negative numbers to binary.

1. Convert the modulus of the number to binary (i.e. drop -ive sign) to binary form.
2. Invert the binary form resulting from step-1. [Replace 0 with 1 and 1 with 0] (aka 1's Complement)
3. Add 1 to the result obtained in step-2 . (Binary addition)

$$(-123)_{10} = (?)_2$$

Steps :

Step-1: Convert 123 to Binary	0 1 1 1 1 0 1 1	We have done it earlier
Step-2: Invert	1 0 0 0 0 1 0 0	
Step-3 Add 1 (binary addition)	Binary 1 : 00000001[8bit] 1 0 0 0 0 1 0 0 0 0 0 0 0 0 0 1 1 0 0 0 0 1 0 1	$1 + 0 = 1$ $0 + 1 = 1$ $1 + 1 = 10$ [2 written as 1 0] $0 + 0 = 0$

$$(-123)_{10} = (10000101)_2 \text{ [2's complement form]}$$

Check

Let us perform operations $123 - 123$

$123 + (-123)$

8 bit format:

1	1	1	1	1	1	1	1	
0	1	1	1	1	0	1	1	123
1	0	0	0	0	1	0	1	-123
0	0	0	0	0	0	0	0	+

Since our representation is in 8-bit format only, so this bit is discarded.

Thus, $123 - 123$ results in:

0	0	0	0	0	0	0	0
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which is 0.